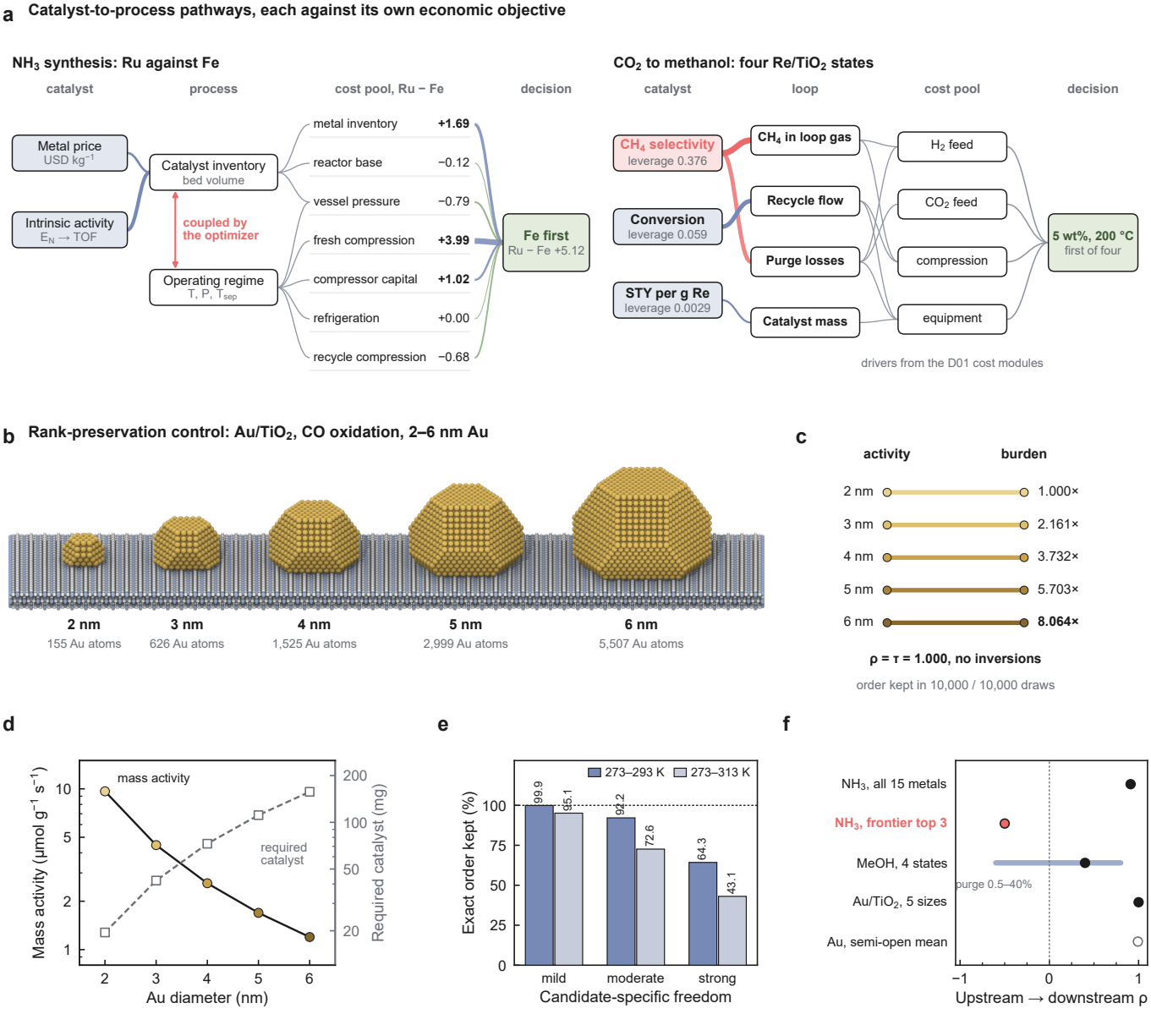

a
Catalyst-to-process pathways, each against its own economic objective
NH3 synthesis: Ru against Fe
catalyst
process
cost pool, Ru − Fe
decision
Metal price
USD kg−1
Intrinsic activity
EN → TOF
Catalyst inventory
bed volume
coupled by
the optimizer
Operating regime
T, P, Tsep
metal inventory +1.69
reactor base −0.12
vessel pressure −0.79
fresh compression +3.99
compressor capital +1.02
refrigeration +0.00
recycle compression −0.68
Fe first
Ru − Fe +5.12
CO2 to methanol: four Re/TiO2 states
catalyst
loop
cost pool
decision
CH4 selectivity
leverage 0.376
Conversion
leverage 0.059
STY per g Re
leverage 0.0029
CH4 in loop gas
Recycle flow
Purge losses
Catalyst mass
H2 feed
CO2 feed
compression
equipment
5 wt%, 200 °C
first of four
drivers from the D01 cost modules
b
Rank-preservation control: Au/TiO2, CO oxidation, 2–6 nm Au
2 nm
155 Au atoms
3 nm
626 Au atoms
4 nm
1,525 Au atoms
5 nm
2,999 Au atoms
6 nm
5,507 Au atoms
c
activity
burden
2 nm 1.000×
3 nm 2.161×
4 nm 3.732×
5 nm 5.703×
6 nm 8.064×
ρ = τ = 1.000, no inversions
order kept in 10,000 / 10,000 draws
d
mass activity
required
catalyst
Mass activity (μmol g−1 s−1)
Required catalyst (mg)
Au diameter (nm)
e
273–293 K
273–313 K
99.9
95.1
94.2
72.6
64.3
43.1
Exact order kept (%)
mild
moderate
strong
Candidate-specific freedom
f
NH3, all 15 metals
NH3, frontier top 3
MeOH, 4 states
purge 0.5–40%
Au/TiO2, 5 sizes
Au, semi-open mean
Upstream → downstream ρ